SOLUTION KEY - MATH 103-113 FINAL

15pt 1) (a) Find the derivative of $f(x) = \arcsin(2x) - x\arctan\left(\frac{x}{2}\right)$

Solution: $f'(x) = \dfrac{2}{\sqrt{1-4x^2}} - \left[1.\arctan\left(\frac{x}{2}\right) + x.\dfrac{1/2}{1+\frac{x^2}{4}}\right]$

(5pt)

$$= \frac{2}{\sqrt{1-4x^2}} - \arctan\left(\frac{x}{2}\right) - \frac{2x}{4+x^2}$$

(b) Find the derivative of $g(x) = 3^x + x^3 + 3^3 + (3x)^{3x}$

Solution: $g'(x) = 3^x.\ln 3 + 3x^2 + 0 + \dfrac{d}{dx}\left((3x)^{3x}\right)$

put $y = (3x)^{3x}$ $\quad \ln y = 3x.\ln 3x$

(5pt)

$$\frac{y'}{y} = 3.\ln 3x + 3x.\frac{1}{x} = 3(\ln 3x + 1)$$

$$y' = (3x)^{3x}.3(\ln 3x + 1)$$

$$\Rightarrow g'(x) = 3^x \ln 3 + 3x^2 + (3x)^{3x}.3.(\ln 3x + 1)$$

(c) If $y = A\sin(\ln x) + B\cos(\ln x)$ where A and B are constants, show that

$$x^2y'' + xy' + y = 0$$

Solution: $y = A\sin(\ln x) + B\cos(\ln x)$

(5pt) $y' = A\cos(\ln x).\frac{1}{x} - B\sin(\ln x).\frac{1}{x}$

$$y'' = -A\sin(\ln x).\frac{1}{x^2} - A\cos(\ln x)\frac{1}{x^2} - B\cos(\ln x)\frac{1}{x^2} + B\sin(\ln x).\frac{1}{x^2}$$

ow,

$$x^2y'' + xy' + y = -A\cancel{\sin(\ln x)} - A\cancel{\cos(\ln x)} - B\cancel{\cos(\ln x)} + B\cancel{\sin(\ln x)} + A\cancel{\cos(\ln x)}$$
$$- B\cancel{\sin(\ln x)} + A\cancel{\sin(\ln x)} + B\cancel{\cos(\ln x)} = 0$$

2) Suppose that y is defined implicitly by the equation
(10pt)
$$(x-1)^2 + y^3 \cos x - 2 = 0$$

(a) Find the derivative of $y(x)$ at the point $(0,1)$.
(b) Find the linearization of $y(x)$ at $x=0$.

Solution: (a) $2(x-1) + 3y^2 \cdot y' \cos x - y^3 \sin x = 0$

(5pt) $\Rightarrow y' = \dfrac{y^3 \sin x - 2x + 2}{3y^2 \cos x}$

$$\boxed{y'\big|_{(0,1)} = \frac{2}{3}}$$

(b) $L(x) = y(0) + y'(0) \cdot (x-0)$

(5pt) $= \boxed{1 + \frac{2}{3}x}$

3) Use the Mean Value Theorem to show that $|\cos x - 1| \leq |x|$ for all
(5pt) x-values. (Hint: Consider $f(t) = \cos t$ on $[0, x]$)

<u>Solution</u>: If we take $f(t) = \cos t$ on $[0, x]$ then it's a continuous function on $[0, x]$ and differentiable on $(0, x)$ for any x-value so we can use Mean Value Theorem to say that there is a $c \in (0, x)$

so that $\dfrac{f(x) - f(0)}{x - 0} = f'(c)$ for any x-value.

$$\Rightarrow \left| \frac{\cos x - \cos 0}{x} \right| = |\sin c| \leq 1 \Rightarrow |\cos x - 1| \leq |x| \text{ for any } x.$$

4) Find $f(4)$ if $\int_0 f(t)\,dt = x.\cos(\pi x)$

(5pt)

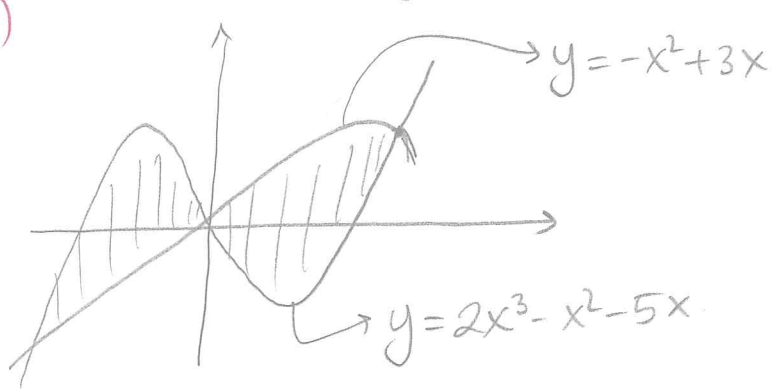
<u>Solution</u>: $\frac{d}{dx}\int_0^x f(t)dt = \frac{d}{dx}(x.\cos(\pi x))$

$\Rightarrow \quad f(x) = 1.\cos(\pi x) - x\sin(\pi x).\pi = \cos(\pi x) - \pi x\sin(\pi x)$

$\Rightarrow f(4) = \cos(4\pi) - \pi.4\sin(4\pi) = 1$ //

5) Find the area of the shaded region.

(12pt)



Solution: $2x^3-x^2-5x=-x^2+3x$

$\Rightarrow 2x^3-8x=0$

(3pt) $\Rightarrow x^3-4x=0$

$\Rightarrow x(x^2-4)=0 \Rightarrow x(x-2)(x+2)=0$

$\Rightarrow x=-2,\ x=0,\ x=2$ are intersection points.

$$A=\int_{-2}^{0} 2x^3-x^2-5x-(-x^2+3x)\,dx + \int_{0}^{2} -x^2+3x-2x^3+x^2+5x\,dx$$

(3pt) (3pt)

$$=\int_{-2}^{0} 2x^3-8x\,dx + \int_{0}^{2} -2x^3+8x\,dx = \frac{x^4}{2}-4x^2\Big|_{-2}^{0} + \frac{-x^4}{2}+4x^2\Big|_{0}^{2}$$

$$=\left[-(8-16)\right]+\left[-8+16\right]$$

$$=8+8=16 \text{ //.}$$ (3pt)

b) Order the following functions from slowest growing to fastest growing as $x \to \infty$

(12pt)

$$(\ln x)^2, x, \sqrt{x}\ln x$$

Solution: $\lim_{x\to\infty} \frac{(\ln x)^2}{x} = \lim_{x\to\infty} \frac{2\ln x \cdot 1/x}{1} = \lim_{x\to\infty} \frac{2\ln x}{x} = \lim_{x\to\infty} \frac{2}{x} = 0.$

(4pt) $(\ln x)^2$ grows slower than x.

$$\lim_{x\to\infty} \frac{\sqrt{x}\ln x}{x} = \lim_{x\to\infty} \frac{\ln x}{\sqrt{x}} = \lim_{x\to\infty} \frac{1/x}{\frac{1}{2\sqrt{x}}} = \lim_{x\to\infty} \frac{2}{\sqrt{x}} = 0.$$

(4pt) $\sqrt{x}\ln x$ grows slower than x

$$\lim_{x\to\infty} \frac{(\ln x)^2}{\sqrt{x}\ln x} = \lim_{x\to\infty} \frac{\ln x}{\sqrt{x}} = \lim_{x\to\infty} \frac{1/x}{\frac{1}{2\sqrt{x}}} = 0.$$

(4pt) $(\ln x)^2$ grows slower than $\sqrt{x}\ln x$

Hence the order from slowest to fastest is

$$\boxed{(\ln x)^2, \sqrt{x}\ln x, x}$$

4) Evaluate the following integrals.

(15pt)

(a) $\int \frac{x+1}{x^4+x^2}\,dx$

Solution: $\frac{x+1}{x^2(x^2+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1}$

$Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2 = x+1$

$A + C = 0$

$B + D = 0$

(6pt) $A = 1 \quad C = -1$

$B = 1 \quad D = -1$

$$\int \frac{x+1}{x^4+x^2}\,dx = \int \frac{1}{x} + \frac{1}{x^2} - \frac{x}{x^2+1} - \frac{1}{x^2+1}\,dx$$

$$= \ln|x| - \frac{1}{x} - \frac{1}{2}\ln|x^2+1| - \arctan x + C.$$

(b) $\int \frac{x^2}{\sqrt{1-4x^2}}\,dx$

Solution: $\int \frac{x^2\,dx}{2\sqrt{\frac{1}{4}-x^2}} = \frac{1}{2}\int \frac{x^2\,dx}{\sqrt{(\frac{1}{2})^2 - x^2}}$

put $x = \frac{1}{2}\sin\theta$

(6pt) $dx = \frac{1}{2}\cos\theta\,d\theta$

$$\int \frac{x^2\,dx}{\sqrt{(\frac{1}{2})^2 - x^2}} = \int \frac{\frac{1}{4}\sin^2\theta \cdot \frac{1}{2}\cancel{\cos\theta}\,d\theta}{\frac{1}{2}\cancel{\cos\theta}} = \frac{1}{4}\int \sin^2\theta\,d\theta = \frac{1}{4}\int \frac{1-\cos 2\theta}{2}\,d\theta$$

$$= \frac{\theta}{8} - \frac{\sin 2\theta}{16}$$

so $\int \frac{x^2\,dx}{\sqrt{1-4x^2}} = \frac{1}{2}\left(\frac{\theta}{8} - \frac{\sin 2\theta}{16}\right) + C$

$$= \frac{\theta}{16} - \frac{\sin 2\theta}{32} + C = \boxed{\frac{\arcsin 2x}{16} - \frac{1}{16}\cdot 2x\cdot\sqrt{1-4x^2} + C.}$$

1(c) $\int xe^{3x}dx$

<u>Solution</u>: put $u = x$ $\quad dv = e^{3x}dx$

(3pt) $\quad du = dx \quad V = \int e^{3x}dx = \frac{1}{3}e^{3x}$

$$\int xe^{3x}dx = \frac{x \cdot e^{3x}}{3} - \frac{1}{3}\int e^{3x}dx = \boxed{\frac{xe^{3x}}{3} - \frac{e^{3x}}{9}}$$

(10pt)

(a) Find the equation of the straight line passing through the points $(0,h)$ and $(r,0)$.

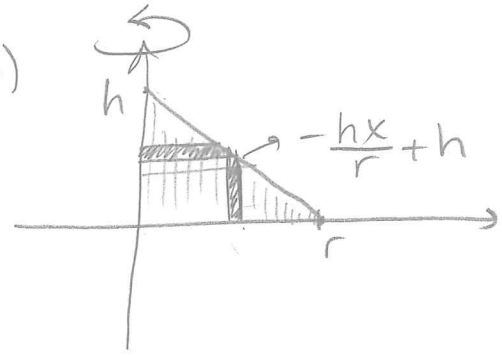
(b) Find the volume of the solid which is generated by revolving the triangular region in the first quadrant whose vertices are $(0,0)$, $(r,0)$ and $(0,h)$ about the y-axis.

Solution: (a) $m_\ell = \dfrac{0-h}{r-0} = -\dfrac{h}{r}$

(3pt) Equation of the line $y - 0 = -\dfrac{h}{r}(x-r)$

$$y = -\frac{hx}{r} + h$$

(b)



Disk Method

(7pt)

$$V = \int_0^h \pi\left(r - \frac{ry}{h}\right)^2 dy$$

$$= \pi \int_0^h r^2 - \frac{2r^2y}{h} + \frac{r^2y^2}{h^2}\, dy$$

$$= \pi\left[r^2y - \frac{r^2y^2}{h} + \frac{r^2}{h^2}\cdot\frac{y^3}{3}\Big|_0^h\right] = \frac{\pi r^2 h}{3} \;//$$

(Alternative Method)

Shell Method

$$V = 2\pi \int_0^r x\cdot\left(-\frac{hx}{r}+h\right)dx$$

$$= 2\pi \int_0^r xh - \frac{hx^2}{r}\, dx$$

$$= 2\pi\left[\frac{x^2h}{2} - \frac{hx^3}{3r}\Big|_0^r\right]$$

$$= 2\pi\left[\frac{r^2h}{2} - \frac{r^2h}{3}\right]$$

$$= 2\pi\cdot\frac{r^2h}{6} = \frac{\pi r^2 h}{3} \;//$$

Note: Students who recognize that the solid is a cone and who write the formula of the volume of the cone will be given only 4 points.

(16pt) Use integration, the Direct Comparison Test or the Limit Comparison Test to test the following integrals for convergence or divergence.

(a) $\int_0^1 -\ln x \, dx$ (b) $\int_1^\infty \frac{dx}{\sqrt{x^6+1}}$ (c) $\int_1^\infty e^{\sqrt{t}} dt$ (d) $\int_2^\infty \frac{1}{\ln x} dx$

(Hint: Use the inequality $\ln x < x$ for all $x \geq 2$)

Solution: (a) $\int_0^1 -\ln x \, dx = \lim_{a\to 0} \int_a^1 -\ln x \, dx = \lim_{a\to 0} \left(x - x\ln x \Big|_a^1 \right)$

(4pt)

$= \lim_{a\to 0} (1 - a + a\ln a)$

$= 1 + \lim_{a\to 0} a \cdot \ln a = 1 + \lim_{a\to 0} \frac{\ln a}{\frac{1}{a}}$

$= 1 + \lim_{a\to 0} \frac{\frac{1}{a}}{-\frac{1}{a^2}} = 1$ //.

Since the result is a finite number it's <u>convergent</u>.

(b) $\frac{1}{\sqrt{x^6+1}} \leq \frac{1}{\sqrt{x^6}} = \frac{1}{x^3}$ and $\int_1^\infty \frac{1}{x^3} dx$ is convergent by p-integra ($p = 3 > 1$).

(4pt)

so by Direct Comparison Test $\int_1^\infty \frac{1}{\sqrt{x^6+1}} dx$ is also convergent.

.) $e^{\sqrt{t}} \geq e$ for $t \geq 1$ and $\int_1^\infty e^{\sqrt{t}} dt \geq \int_1^\infty e \, dt = e \int_1^\infty dt$

(4pt)

and $\int_1^\infty dt = \lim_{b\to\infty} \int_1^b dt = \lim_{b\to\infty} t \Big|_1^b = \lim_{b\to\infty} b - 1 = \infty$.

so $\int_1^\infty e \, dt$ is divergent and by Direct Comparison Test $\int_1^\infty e^{\sqrt{t}} dt$ is also divergent.

1) $\ln x < x \Rightarrow \frac{1}{\ln x} > \frac{1}{x}$ and $\int_2^\infty \frac{1}{x} dx$ is divergent by p-integral ($p = 1$)

(4pt)

o by Direct Comparison Test $\int_2^\infty \frac{1}{\ln x} dx$ is also divergent.